MILITARY SPECIFICATION

PROTECTOR, DENTURE MODEL, FOAMED PLASTIC, THREE-PIECE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a foamed plastic protector suitable for the receipt and protection of dental models and appliances.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-636	-	Boxes, Shipping, Fiberboard.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 S. 20th Street, Phila., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D1564 - Standard Methods of Testing Flexible Cellular Materials - Slab Urethane Foam

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

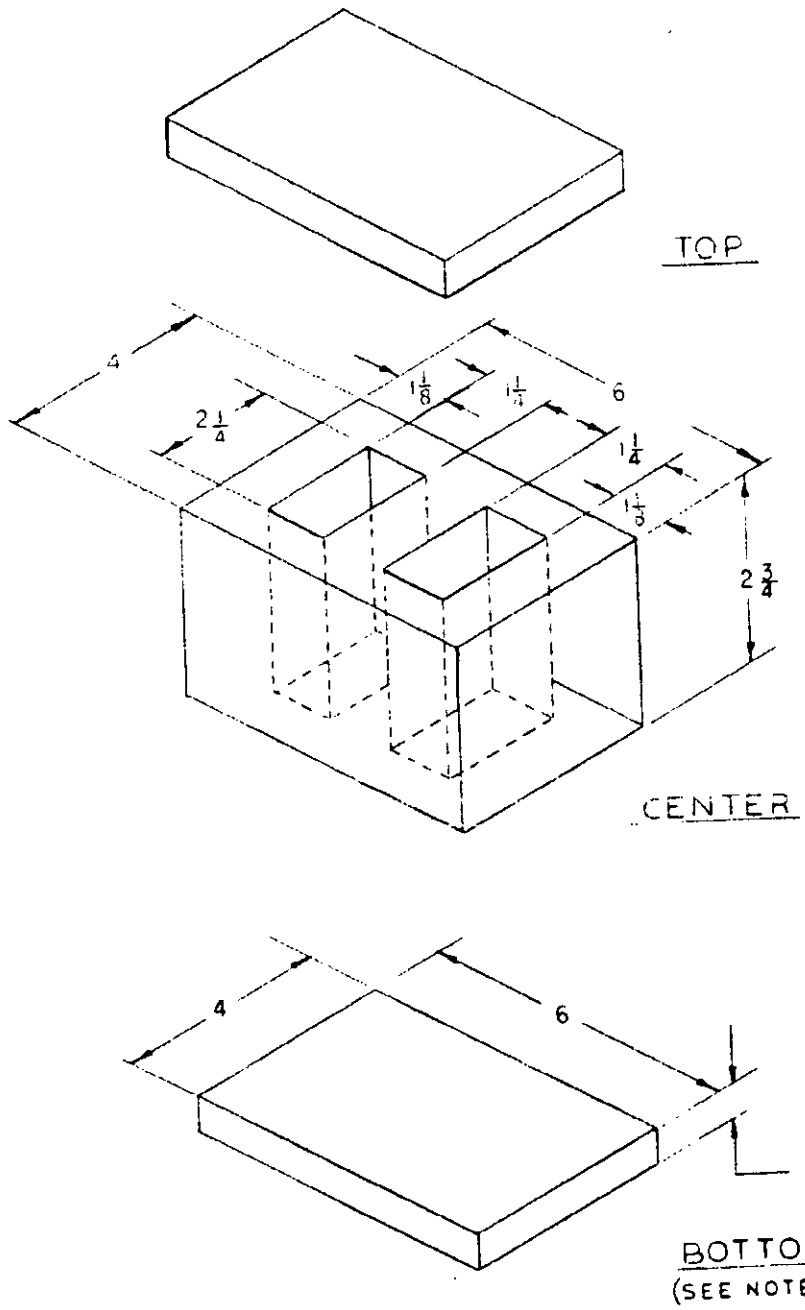
3. REQUIREMENTS

3.1 Material. The protector shall be fabricated from high quality polyurethane foam which shall be entirely suitable for the intended purpose, and which shall provide a product conforming to the requirements specified herein.

3.2 Style, design, and construction. The protector shall be of three-piece construction consisting of a top piece, bottom piece, and center piece, all separate and preformed. The top and bottom pieces shall be preformed into the shapes of rectangular blocks. The center piece shall be preformed into the shape of a rectangular block with two rectangular shaped holes through the entire thickness. The protector shall be a device suitable for repeated use in protecting dental models and appliances used in the construction of restorative prostheses during shipment of these models and appliances between laboratories. The protector shall conform to figure 1 for style, design, and dimensions.

3.2.1 Figures. The figures forming a part of this specification are not production figures. Figures establish dimensional limitations and control design and configuration. The supplier shall be responsible for preparing his own shop drawings. The supplier shall select dimensions and tolerances within the limitations prescribed on the figures to provide a product with proper design, configuration, fit, and function. No deviations from the dimensional limitations prescribed on the figures are permissible without prior approval of the contracting officer.

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NOTE:
DIMENSIONS FOR TOP SAME AS THOSE SHOWN FOR BOTTOM
Dimensions are in inches
Unless otherwise specified, tolerances shall be as follows:

Dimensions 2 in. and aboye ± 1/4
Dimensions under 2 in. ± 1/8

FIGURE 1. Protector, Denture Model.

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3.3 Physical properties. Polyurethane foam shall conform to the physical properties listed in table I, when tested as specified in 4.3.

TABLE I. Physical properties.

Characteristic	Requirement
Constant deflection compression set	10 percent, maximum
Indentation load deflection	28 to 35 pounds
Density	1.20 ± 0.05 pounds per cubic foot
Tensile strength	10.0 pounds per square inch, minimum
Elongation	150 percent, minimum
Resilience	40 percent, minimum

3.4 Odor. The protector shall be free from any objectionable odor.

3.5 Workmanship. Workmanship shall be first class throughout. Protectors shall be free from defects which may detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Certificates of quality. Certificates of quality, supplied by the manufacturer of the polyurethane foam material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificates shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties). Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Examination. Protectors shall be examined for defects, including, but not limited to, those listed in table II.

TABLE II. Classification of defects.

Categories	Defects
Major A:	
101	Style and design of protector not as specified.
102	Protector not of three-piece construction.
103	Protector not free from unblown resin area.
104	Large tears or rips present in any of component pieces of protector.
105	Objectionable odor present.
Major B:	
151	Dimensions of any of component pieces of protector not as specified.
Minor:	
201	Number of protectors in unit package not as specified.

The inspection levels and acceptable quality levels (AQL's) shall be as listed in table III. The unit of product for examination shall be one protector.

TABLE III. Examination criteria.

Type defect	Inspection level	AQL (Percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor	II	2.5

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4.2.2 Testing. Protectors shall be tested as specified in table IV and 4.3. For tests conducted on raw material, a sufficient quantity of each lot of raw material (polyurethane foam) shall be taken for the sample unit. Number of samples to be taken and dimensions of samples shall conform to requirements specified in the American Society for Testing and Materials (ASTM) Method D1564. As an alternate, the Constant Deflection Compression Set, Density, and Tension tests may be run on samples taken from each lot of the finished product (protector), provided that dimensions of sample and number of samples to be taken conform to requirements specified in ASTM Method D1564.

4.2.3 Packaging inspection. The inspection of the unit package, packing, and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.3 Tests. Polyurethane foam shall be tested for the physical properties listed in 3.3 in accordance with the applicable ASTM test methods as shown in table IV.

TABLE IV. Test methods (ASTM D1564).

Constant Deflection Compression Set test
Load Deflection test. Method A, Indentation to specified deflections (25 percent on 4 inches)
Density test
Tension test (for tensile strength and elongation)
Resilience test

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One package containing one hundred protectors (three-piece), as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. One hundred protectors (top, bottom and center section) shall be neatly arranged and packaged in a box conforming to PPP-B-636, type CF, class domestic. When protectors are stacked in two or more tiers, a fiberboard pad shall be interposed between each tier. Closure of the box shall be as specified in the appendix of the box specification.

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5.1.3 Level C. One hundred protectors (top, bottom and center section) shall be clean, dry and packaged in a manner that will afford adequate protection against physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Units packaged as specified in 5.1.2.1 shall require no further packing except that the unit package shall conform to PPP-B-636, class weather-resistant. In addition to the closure, waterproofing and strapping shall apply as specified in the appendix of the box specification.

5.2.2 Level B. Units packaged as specified in 5.1.2.1 shall require no further packing.

5.2.3 Level C. Protectors packaged as specified in 5.1.3 shall require no further packing providing these packs conform to the applicable carrier rules and regulations.

5.3 Marking. Marking shall conform to the requirements as specified in MIL-STD-129. When the unit package is utilized as the exterior container, the exterior container markings as specified in MIL-STD-129 shall apply. The level of protection shall be shown as "A", "B" or "C" as applicable in lieu of "A/A", "A/B" or "C/C".

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

NSN	Item Identification
6520-00-142-8727	PROTECTOR, DENTURE MODEL, Foamed Plastic, Three-piece, 100s.

6.3 Supersession data. This specification includes the requirements of Purchase Description No. 1 dated 11 June 1969.

Military Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA - DM

Project No. 6520-1423

Review activity:
DOD - MB

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers, vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-P-37500 - PROTECTOR, DENTURE MODEL, FOAMED PLASTIC, THREE-PIECE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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PREVIOUS EDITION OF 1 JAN 68 WHICH MAY BE USED

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